

6. Write a C program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U.” Break this code.

Aim:

To write a C program to break an Affine cipher using frequency analysis, where the most frequent ciphertext letter is B and the second most frequent letter is U.

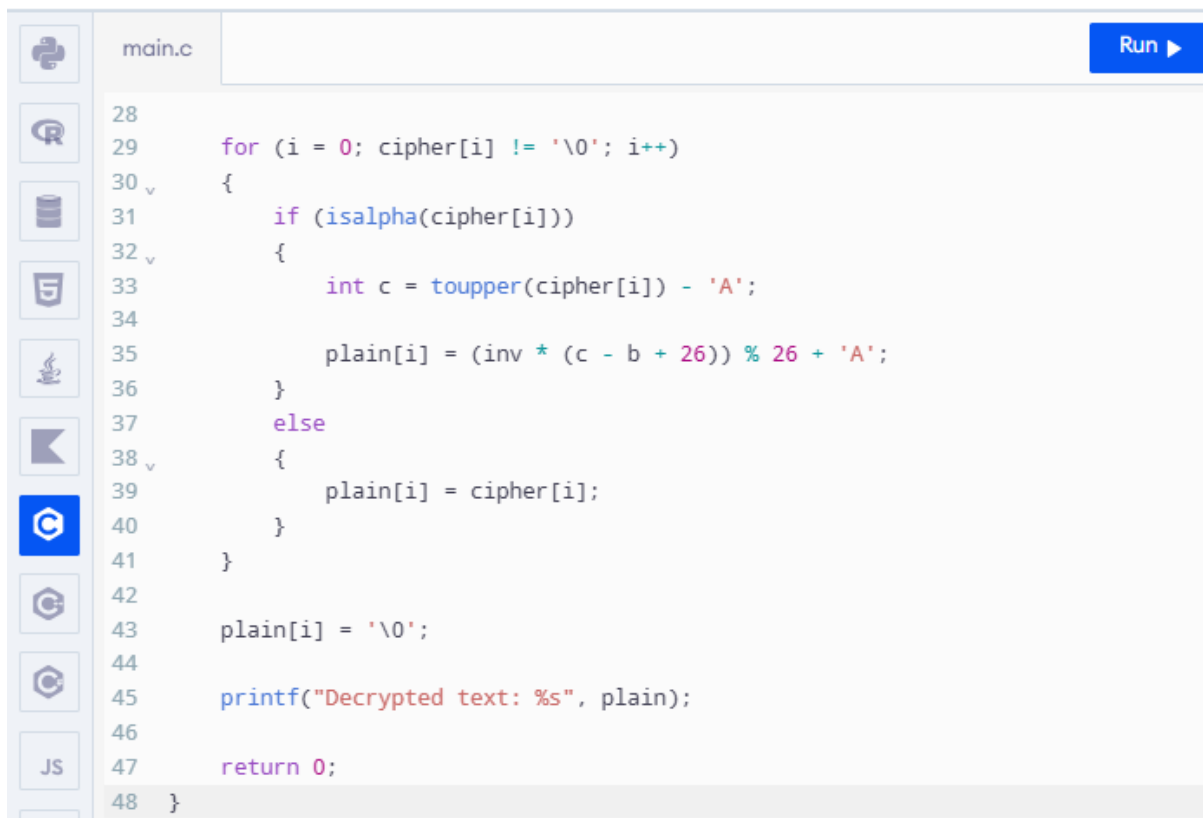
Algorithm:

1. Start the program.
2. Assume the most frequent English letter E maps to ciphertext B.
3. Assume the second most frequent English letter T maps to ciphertext U.
4. Use the equations $4a + b \equiv 1 \pmod{26}$ and $19a + b \equiv 20 \pmod{26}$.
5. Find the values of a and b.
6. Calculate the multiplicative inverse of a modulo 26.
7. Use the inverse to decrypt the ciphertext and display the plaintext.

Program:



```
main.c Run ▶  
1  #include <stdio.h>  
2  #include <string.h>  
3  #include <ctype.h>  
4  
5  int inverse(int a)  
6  {  
7      int i;  
8  
9      for (i = 1; i < 26; i++)  
10     {  
11         if ((a * i) % 26 == 1)  
12             return i;  
13     }  
14  
15     return -1;  
16 }  
17  
18 int main()  
19 {  
20     char cipher[200], plain[200];  
21     int i;
```



```
28
29     for (i = 0; cipher[i] != '\0'; i++)
30     {
31         if (isalpha(cipher[i]))
32         {
33             int c = toupper(cipher[i]) - 'A';
34
35             plain[i] = (inv * (c - b + 26)) % 26 + 'A';
36         }
37         else
38         {
39             plain[i] = cipher[i];
40         }
41     }
42
43     plain[i] = '\0';
44
45     printf("Decrypted text: %s", plain);
46
47     return 0;
48 }
```

Output:

Output

```
Enter ciphertext: B U
Decrypted text: U L
```

Result:

Thus, the Affine cipher was successfully broken using frequency analysis.